Utkarsh Singh

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Education

Vellore Institute of Technology, Bhopal 2023 – 2027
Bachelor of Technology in Computer Science ***CGPA: 8.63***

Pt. Deen Dayal Upadhyaya Sanatan Dharma Vidyalaya, Kanpur, U.P. 2021 – 2022
Class XII ***CBSE: 86.8***

Maharishi Vidya Mandir, Fatehpur, U.P. 2019 – 2020
Class X ***CBSE: 93.6***

Technical Skills

Programming Languages: Java, Python, C++, JavaScript, TypeScript

Backend: Node.js, Express.js, REST APIs

Databases: MongoDB, MySQL

Frontend: HTML, CSS, Tailwind CSS, React.js, Next.js

AI/ML: PyTorch, NumPy, Pandas, OpenCV

Developer Tools: Git, GitHub, Postman, AWS, Figma, MATLAB, Jupyter Notebook

Concepts: Computer Vision, Image Augmentation, Transfer Learning, CNN, Vision Transformers

Projects

Full-Stack Q&A Platform Next.js 14, Appwrite, TypeScript, Zustand, Tailwind CSS
Full-Stack Project

- Developed a StackOverflow-style Q&A platform with user authentication, question posting, answering, and voting using **Appwrite** as backend-as-a-service.
- Configured **Next.js middleware** to auto-initialize Appwrite database and storage, eliminating manual backend setup.
- Designed a **reputation system** with dynamic score updates and global auth state management using **Zustand with Immer**.

Real-Time Collaborative Spreadsheet Next.js, TypeScript, Tailwind CSS, Firebase
Frontend Project

- Constructed a real-time collaborative spreadsheet with multi-user editing and live synchronization using Firebase.
- Architected a responsive grid with formula support and keyboard navigation, increasing data entry speed by 25%.
- Integrated user presence tracking and identity system for collaborative awareness.

DB-ATP: Crop Disease Diagnosis Model Python, PyTorch, MobileNetV3, DeiT, OpenCV
ML Research Project — Conference Submission

- Co-developed a Dual-Branched Hybrid CNN-Transformer model for multi-crop disease classification, achieving **87.8% accuracy** and **90.2% precision** across wheat and soybean datasets.
- Engineered an Adaptive Token Pruning (ATP) module that dynamically discards 40–50% of background tokens, reducing inference latency to **7.3 ms** for real-time edge deployment.
- Implemented Multi-Head Cross-Attention fusion to combine CNN texture features with Transformer global context, outperforming ResNet-50, EfficientNet-B0, and ViT-B/16 baselines.

Positions of Responsibility

Core Member — Design Team (MUN Club) VIT Bhopal
Design & Visual Communications *2024 – Present*

Designed posters, social media creatives, and branding assets for Model United Nations conferences and promotional campaigns.

Collaborated with organizing and outreach teams to ensure consistent visual identity across all event materials.

Certifications

- Google: The Bits and Bytes of Computer Networking
- NPTEL: Introduction to Machine Learning
- Great Learning Academy: Linux Tutorial